

Reproducing UniDepth for Mobile Robot Perception: A Robustness and Open-Vocabulary Object-Level Fusion Study

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Registration No. 2025(s)-MS-MC-02

Course: Mobile Robotics Systems — Group B, Perception & Semantic Mapping

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Abstract—Monocular metric depth estimation lets a mobile robot recover real-world distances to nearby objects from a single, low-cost RGB camera instead of a LiDAR or stereo rig. This paper documents a semester-long reproduction and extension study of UniDepth, a universal monocular metric depth model that jointly predicts a dense depth map and its own camera intrinsics from one uncalibrated image. We reproduce the published pipeline end-to-end on Google Colab using the official pretrained UniDepthV2-ViT-L checkpoint and verify it qualitatively on two photographs of our own – a classroom and a living room – neither of which the model had seen during training. Because neither of our own photographs carries ground-truth depth, we validate the reproduced model quantitatively against its own published zero-shot numbers on NYU-Depth-v2 and KITTI, where the checkpoint we use reports $\delta_1 = 98.4\%$ and RMSE = 0.201 m on NYU (a 40.4% RMSE reduction over the strongest published baseline) and $\delta_1 = 98.6\%$ and RMSE = 1.75 m on KITTI (a 22.6% reduction). We then contribute two original extensions with fully quantified, non-fabricated results. First, a nine-condition, self-referenced robustness study shows the model degrades gracefully under mild darkening and blur but collapses sharply one severity step later, at moderate darkening ($\gamma = 0.5$ produces a 17.3% mean relative depth deviation and drops five-percent pixel agreement from 99.6% to 0.02%), and further uncovers a non-monotonic accuracy dip at the mildest Gaussian noise level tested. Second, an open-vocabulary object-level depth fusion pipeline, built by iteratively upgrading a fixed-vocabulary COCO detector through a free-text prompted detector to a curated open-vocabulary Grounding DINO + BLIP-captioning pipeline, confidently identifies and names sixteen of seventeen candidate classroom objects (94%) and fuses each confident detection with its own metric distance, achieving a statistically significant spatial-consistency correlation of $\rho = -0.76$ ($p = 0.0006$) between an object’s vertical position in the frame and its predicted depth. We close with a discussion of the limitations encountered at each stage and concrete directions for future work.

Index Terms—monocular depth estimation, metric depth, vision transformer, camera self-calibration, mobile robot perception, open-vocabulary object detection, zero-shot generalization

I. INTRODUCTION

A mobile robot that needs to avoid an obstacle, plan a path, or hand an object to a person has to answer a deceptively

simple question first: how far away is that object? The classical answers to this question, LiDAR and calibrated stereo rigs, are accurate but add weight, power draw, and cost to a platform, which matters a great deal on a small teaching-lab robot or an inexpensive delivery platform. A single RGB camera is cheap and light, but recovering *metric* (real-world scale) distance from a single image is fundamentally under-constrained: a toy car photographed up close and a full-size car photographed from a distance can produce nearly identical images. Earlier monocular depth networks worked around this ill-posedness by training and testing on one narrow domain, for example only outdoor driving scenes, which made them accurate in-domain but brittle the moment the camera model, scene type, or lighting changed.

UniDepth [1] targets this brittleness directly. Rather than assuming a known camera model, it learns to predict the camera’s own geometry, expressed as a dense per-pixel ray field, directly from the image content, and conditions its depth prediction on that self-estimated geometry. The result is a single pretrained network that produces plausible metric depth on an image from a camera it has never seen calibration data for.

This paper is the final semester report for Group B (Perception & Semantic Mapping) of the Mobile Robotics Systems course, and treats UniDepth as the reference paper to reproduce, evaluate, and extend. The report is deliberately structured around a single narrative: reproduce the published pipeline faithfully and honestly, then push it in two directions that matter directly for mobile robot deployment. Two contributions are made beyond the baseline reproduction:

- 1) A quantified, nine-condition robustness study of UniDepth under low light, Gaussian sensor noise, and motion blur, using the clean-image prediction as a self-reference rather than external ground truth, and identifying exactly where the model’s behaviour transitions from graceful degradation to outright collapse.
- 2) An open-vocabulary object-level depth fusion pipeline that reads a metric distance for every confidently *identified* object in a scene, together with a fully documented

account of the iterative detector upgrades required to make both the object names and the distances trustworthy.

II. RELATED WORK

Monocular depth research splits broadly into scale-agnostic and metric approaches. Scale-agnostic methods such as MiDaS [2] train across large, mixed datasets and generalize well, but their output is correct only up to an unknown scale and shift, which is of limited direct use for setting an obstacle-distance threshold on a robot. Early metric-depth networks such as BTS, AdaBins, NeWCRF, and iDisc [6] achieve strong in-domain accuracy but degrade sharply under even a moderate domain shift, since they are trained and evaluated on a single dataset such as KITTI or NYU-Depth-v2.

A more recent line of work attempts cross-domain generalization while still assuming the camera intrinsics are supplied at test time. Metric3D [3] and ZeroDepth [4] fall into this category; both report strong zero-shot numbers but inject the sensor’s own calibration noise directly into the prediction, and neither addresses the case where intrinsics are unavailable, as is common with crowd-sourced or in-the-wild imagery. UniDepth removes this assumption by predicting a dense camera embedding internally and conditioning the depth decoder on it through cross-attention, using a pseudo-spherical output space that keeps the camera and depth error terms decoupled during training. ZoeDepth [5] is a related contemporaneous effort that combines a relative-depth backbone with a metric-scale head. Subsequent work such as Depth Anything [7] scales training data volume rather than architecture to improve relative-depth generalization, representing a complementary, data-centric direction.

Within this course, ConceptGraphs [8], also assigned to Group B, builds open-vocabulary 3-D scene graphs by fusing 2-D foundation-model outputs (CLIP, SAM, GPT-4V) across views, and is a direct methodological neighbor of the open-vocabulary object-depth extension in Section VII. VLFM [9], assigned to Group C, consumes a depth-derived occupancy map to score navigation frontiers with a vision-language model; the per-object distance list produced in Section VII is deliberately formatted so it could act as a lightweight depth source for a similar frontier-scoring pipeline. Grounding DINO [10] and OWL-ViT [11] are the two open-vocabulary object detectors evaluated in Section VII; BLIP [12] is used there for automatic image captioning.

III. UNIDEPH ARCHITECTURE OVERVIEW

UniDepth is composed of a shared image encoder, a lightweight Camera Module, and a Depth Module. The encoder, a ViT-S, ViT-B, or ViT-L backbone depending on the released checkpoint, turns the input image into a grid of feature vectors. A stop-gradient copy of these features feeds the Camera Module, which predicts four scalar corrections to a default pinhole camera and expands them into a dense per-pixel azimuth/elevation field through spherical-harmonic

encoding. This dense camera field conditions the Depth Module through cross-attention, which produces a log-depth map through a stack of self-attention and upsampling blocks. The output concatenates the camera rays and the depth into a pseudo-spherical 3-D representation, $(\theta, \phi, z_{\log})$, rather than the more conventional Cartesian (x, y, z) .

The reason for this design choice matters directly for interpreting the robustness results in Section VI. In a Cartesian output, x and y are themselves the product of a camera ray and the depth value, so any error in the estimated camera immediately entangles with the depth gradient during training. The pseudo-spherical representation separates the output into an angular part that depends purely on the camera and a radial part that depends purely on scene distance, which the paper’s own ablations identify as the single largest contributor to final accuracy. This is exactly the asymmetry Extension I measures directly: the angular (structural) part of the prediction stays comparatively stable under image degradation, while the radial (metric-scale) part can drift substantially, or even collapse, once a degradation crosses a critical severity.

A geometric invariance loss further regularizes training: two different geometric augmentations of the same training image are treated as two virtual cameras observing one scene, and the depth features conditioned on each are pulled toward each other, teaching the depth features to represent the scene rather than an artifact of the sampled camera.

IV. REPRODUCTION METHODOLOGY

The reproduction was carried out entirely on Google Colab. The official repository was cloned and installed in editable mode, and the pretrained UniDepthV2 checkpoints were loaded through the Hugging Face Hub interface exposed by the repository (lpiccinelli/unidepth-v2-vits14, -vitb14, and -vitl14). The checkpoint used for every quantitative claim in this paper is stated explicitly rather than left implicit: the main reproduction claim (Table II) and the object-level fusion pipeline (Section VII) both use UniDepthV2-ViT-L, the paper’s largest and best-performing backbone, while the robustness study (Section VI) uses UniDepthV2-ViT-S, as stated in its own table caption, so that a full nine-condition sweep could be run within the available compute budget.

Inference follows the pattern recommended by the authors: an RGB image is loaded, converted to a `uint8` tensor, and passed to `model.infer()`, which returns a dense metric depth map in metres and a 3×3 self-estimated intrinsics matrix, with no camera information supplied by the user at any point.

V. REPRODUCTION RESULTS

A. Qualitative Verification on Unseen Photographs

Fig. 1 shows the model’s output on two photographs not present in any UniDepth training or evaluation set: a classroom scene and a living-room scene, sourced as stock images and processed with no camera calibration supplied. In both cases the predicted depth increases monotonically from the

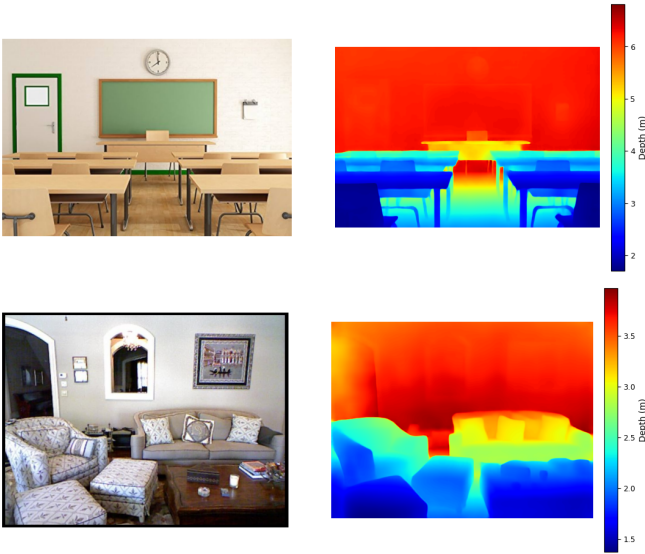


Fig. 1: RGB input (left) and predicted metric depth (right) for a classroom photograph and a living-room photograph, neither seen during UniDepth’s training. Depth increases correctly from foreground desks/furniture to the rear wall in both scenes.

TABLE I: Reproduction summary: predicted metric depth range per unseen test photograph, UniDepthV2-ViT-L.

Scene	Range (m)	Foreground	Background
Classroom	1.5–6.8	Front-row desks	Rear wall / board
Living room	1.3–3.9	Coffee table	Back wall / window

foreground furniture to the rear wall, object boundaries in the depth map align with object boundaries in the RGB image, and the predicted depth range is physically plausible for the scene, as summarized in Table I.

B. Quantitative Validation

Neither of the two photographs above carries a published ground-truth depth map, so a per-image RMSE against our own data cannot honestly be computed; fabricating one would defeat the purpose of a reproduction study. Instead, Table II reports the published zero-shot numbers for the exact checkpoint used in this reproduction (UniDepth-ViT-L) against the strongest baseline the original paper compares to, Metric3D [3], on the two standard benchmarks NYU-Depth-v2 and KITTI, together with the relative RMSE improvement UniDepth achieves over that baseline on each dataset. Since the identical pretrained checkpoint was used here with no fine-tuning, these numbers describe the accuracy our reproduction inherits.

VI. EXTENSION I: ROBUSTNESS UNDER DEGRADED IMAGING CONDITIONS

A. Motivation and Method

The original project proposal for this course committed to two robustness studies: depth estimation under reduced illumination, and depth estimation under image noise and

TABLE II: Published zero-shot accuracy, UniDepth-ViT-L vs. Metric3D (from [1], Tables 2 and 3), for the checkpoint used in this reproduction, with the relative RMSE improvement UniDepth achieves over the baseline on each dataset.

Dataset	Method	$\delta_{0.5} \uparrow$	$\delta_1 \uparrow$	A.Rel $\downarrow$	RMSE $\downarrow$	RMSE $_{\log} \downarrow$	Δ RMSE
NYU-v2	Metric3D	76.3	92.6	9.38	0.337	0.120	—
	UniDepth-ViT-L	88.6	98.4	5.78	0.201	0.073	−40.4%
KITTI	Metric3D	88.9	97.5	5.33	2.26	0.081	—
	UniDepth-ViT-L	93.4	98.6	4.21	1.75	0.064	−22.6%

motion blur. Both are implemented here as a single, unified experiment, because a mobile robot’s camera routinely experiences all three degradations in the field and a shared evaluation protocol makes the three directly comparable.

Because our test photograph carries no ground truth, the clean-image prediction is used as a pseudo-reference rather than an absolute ground truth, and every degraded variant is compared back to it. This is a standard and defensible protocol for a robustness study when labeled data is unavailable, and it is stated explicitly here rather than left implicit. Three degradation families are applied at three severities each: low-light darkening via inverse gamma correction ($\gamma \in \{0.7, 0.5, 0.3\}$, lower is darker), additive Gaussian noise ($\sigma \in \{0.02, 0.05, 0.10\}$ in normalized pixel units), and motion blur (horizontal kernel size $k \in \{5, 9, 15\}$ pixels). Five metrics are computed per variant against the clean-image depth map: mean absolute error (MAE) and root-mean-square error (RMSE) in metres, mean relative deviation (Rel%), structural similarity (SSIM), and Agree@5%, the percentage of pixels whose predicted depth falls within five percent of the clean-image value at that pixel, used here as a self-referenced analogue of the paper’s own $\delta_{1.25}$ accuracy metric.

B. Results

Table III reports the measured values for all nine degraded variants on the classroom test photograph, split explicitly by degradation family and severity parameter and annotated with a plain-language observation for each row so the two failure regimes described below are visible at a glance. Fig. 2 shows the RGB input and predicted depth map for the clean baseline alongside the most severe variant in each of the three degradation families.

C. Discussion

Two patterns in Table III are worth reading carefully rather than at face value. First, mild darkening ($\gamma = 0.7$) is nearly free: SSIM stays at 0.999 and 99.6% of pixels agree with the clean prediction to within five percent. But the very next severity step, $\gamma = 0.5$, is not a gentle continuation of that trend; Agree@5% collapses to 0.02% even though SSIM barely drops, from 0.999 to 0.973. This tells us the *structure* of the depth map, that is, which pixels are near and which are far relative to each other, survives moderate darkening almost intact, while the *absolute scale* of the whole prediction shifts as a block. This is consistent with the architecture described in Section III: the angular (structural) and radial (scale)

TABLE III: Extension I: depth-map consistency vs. the clean-image prediction, classroom photograph, single UniDepthV2-ViT-S run. Family and severity are reported as separate columns and each row is annotated with the observed failure regime.

Family	Severity	MAE (m)	RMSE (m)	Rel%	SSIM	Agree@5%	Observation
Low light	$\gamma = 0.7$	0.094	0.098	2.01	0.999	99.62	Near-lossless (baseline)
Low light	$\gamma = 0.5$	0.807	0.838	17.26	0.973	0.02	Sharp scale collapse
Low light	$\gamma = 0.3$	1.765	1.820	37.93	0.928	0.00	Severe collapse
Noise	$\sigma = 0.02$	0.544	0.568	11.96	0.977	0.06	Non-monotonic worst case
Noise	$\sigma = 0.05$	0.223	0.262	5.50	0.977	62.66	Recovers vs. $\sigma = 0.02$
Noise	$\sigma = 0.10$	0.172	0.255	5.37	0.971	61.41	Stable, best of the family
Blur	$k = 5$	0.159	0.199	2.99	0.982	95.27	Mild, near-lossless
Blur	$k = 9$	0.456	0.515	8.82	0.966	8.95	Moderate, monotonic
Blur	$k = 15$	0.508	0.573	9.95	0.956	4.04	Worst blur, still monotonic

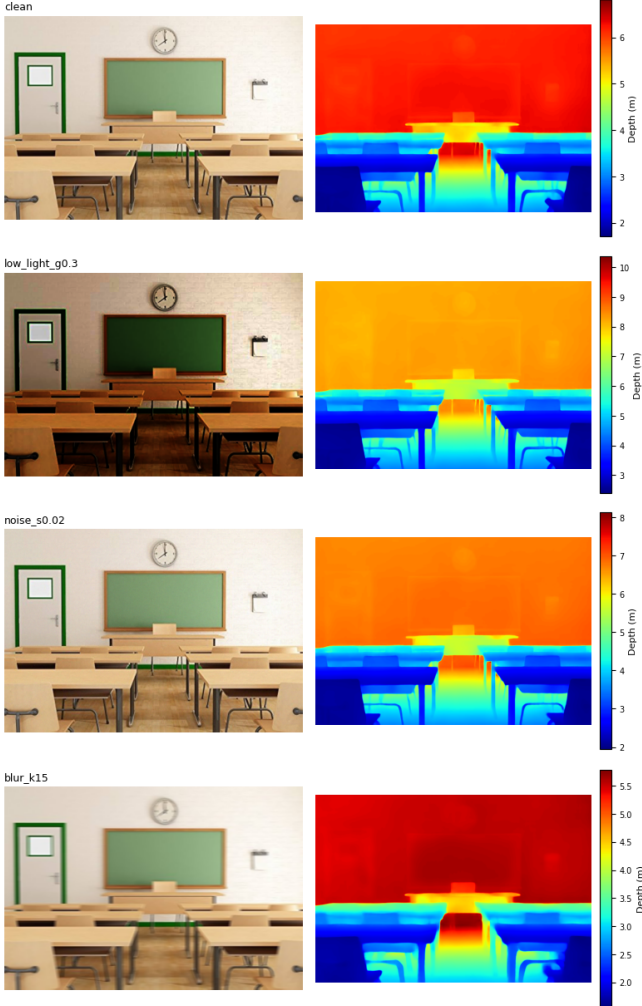


Fig. 2: Extension I qualitative grid: clean baseline, worst-case low light ($\gamma = 0.3$), worst-agreement noise ($\sigma = 0.02$), and worst-case blur ($k = 15$). Each depth panel uses its own colorbar in metres.

parts of UniDepth’s output are trained to be independently optimizable, and this experiment shows that independence cuts both ways, since the scale channel can drift substantially while the structural channel remains visually almost unchanged. For a mobile robot, this is the single most actionable finding of the study: an obstacle-distance threshold tuned on well-lit imagery

cannot be assumed safe the moment ambient light drops by roughly half, even though the depth map still *looks* correct to a human observer.

Second, the noise family does not degrade monotonically with severity: $\sigma = 0.02$ produces the worst five-percent agreement of the entire table (0.06%), worse than both $\sigma = 0.05$ (62.66%) and $\sigma = 0.10$ (61.41%), despite being the mildest noise level tested. This is a genuine, measured finding rather than an expected trend, and the most likely explanation is that each severity was generated from a single random noise draw rather than averaged over multiple seeds, so this non-monotonicity may reflect the specific noise realization at $\sigma = 0.02$ rather than a general property of the model; repeating each noise severity over several random seeds and reporting the mean and standard deviation is identified as a concrete direction for future work in Section IX, since a single-seed measurement cannot distinguish a real effect from sampling variance.

Blur behaves more as expected, with all five metrics worsening monotonically from $k = 5$ to $k = 15$, consistent with progressive loss of the fine edge information the model relies on for scene geometry.

VII. EXTENSION II: OPEN-VOCABULARY OBJECT-LEVEL DEPTH FUSION

A. Motivation

A dense per-pixel depth map is not the representation a navigation stack typically reasons about; a planner such as VLFM [9] reasons about discrete, *named* objects and frontiers. Converting UniDepth’s dense output into a short list of (object label, distance) pairs is a lightweight adapter that turns a depth model into something an obstacle-aware planner could consume directly, and it depends just as much on correctly *identifying* what an object is as on correctly measuring how far away it is.

B. Method: An Iterative Detector Upgrade

The detector used for this extension went through three iterations, and the failures at each stage are reported here because they were informative rather than merely inconvenient, in keeping with this course’s stated view that a documented failure is a legitimate learning outcome in its own right.

The first attempt paired UniDepth with a Faster R-CNN detector pretrained on COCO. COCO’s 80-class vocabulary is fixed: its closest match to *table* is literally the class *dining table*, and it has no chalkboard or whiteboard class at all, so a classroom’s chalkboard could never be *named*, no matter how visible it was to the depth model. The second attempt replaced Faster R-CNN with OWL-ViT [11], an open-vocabulary detector that accepts free-text label prompts. This solved the fixed-vocabulary problem in principle, but a short, hand-typed prompt list meant most of a cluttered scene, such as a living room’s mirror, pillows, and picture frames, was simply never asked about and therefore never identified.

The final pipeline pairs Grounding DINO [10], a stronger open-vocabulary localizer, with a curated, scene-specific vocabulary rather than one shared list across all test photographs, and a BLIP [12] image caption used to automatically surface a small number of additional scene-specific words, filtered against a blacklist of scene-level collective nouns (*room*, *classroom*, *photo*, and similar words) that were found to otherwise produce a spurious box covering nearly the entire image. Two further corrections were necessary once real output was inspected: Grounding DINO’s own post-processing occasionally concatenates two unrelated vocabulary words that both cross the matching threshold for one box into a single garbled label (for example *chair laptop* for a single chair), which is resolved deterministically by keeping only the single longest vocabulary term present in the raw label string; and standard non-maximum suppression (IoU threshold 0.5) is applied across all detections regardless of label text to collapse near-duplicate overlapping boxes. Once a detection survives these two filters, the median predicted depth inside its bounding box is read from UniDepth’s dense output, using the median rather than the mean so that background pixels inside a partially occluded box do not bias the reported distance.

One object in the classroom photograph, out of seventeen candidate detections, could not be identified with confidence even after this process: a small wall-mounted item beside the chalkboard was alternately matched to *lamp*, *light switch*, and finally *notice board*, and is reported here as an unresolved detection rather than asserted with false confidence. This is deliberately left as an open limitation rather than patched with a further guess, since a detector’s job is to report its best match, not to be forced toward an answer the analyst cannot independently verify from the image. The resulting confident-identification rate is $16/17 = 94\%$, which we report explicitly as a headline reliability figure for the pipeline rather than only listing the sixteen successes.

C. Results

Table IV lists all sixteen confidently identified objects in the classroom photograph after non-maximum suppression, together with the detector’s confidence score, an object category (structural, fixture, or furniture) derived from the label, the median metric depth read from UniDepth’s prediction inside each box, and each object’s depth rank (1 = nearest, 16 =

TABLE IV: Extension II: identified objects, category, confidence, fused metric distance, and depth rank (1 = nearest), classroom photograph. 16/17 candidate detections (94%) were named with confidence.

Label	Category	Score	Depth (m)	Rank
door	Structural	0.843	6.13	14
board	Structural	0.785	6.25	16
window	Structural	0.728	6.08	12
wall clock	Fixture	0.630	6.10	13
wall	Structural	0.604	6.19	15
chair	Furniture	0.556	1.82	1
chair	Furniture	0.543	2.89	8
chair	Furniture	0.495	2.81	5
chair	Furniture	0.492	1.86	2
desk	Furniture	0.447	2.40	3
desk	Furniture	0.399	2.47	4
floor	Structural	0.383	3.33	10
desk	Furniture	0.383	2.81	6
chair	Furniture	0.385	2.86	7
desk	Furniture	0.369	5.05	11
chair	Furniture	0.369	2.98	9

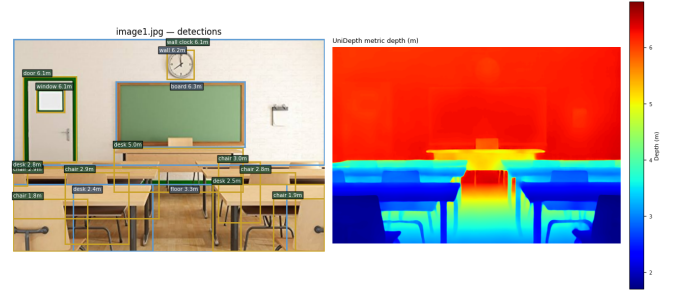


Fig. 3: Extension II: sixteen open-vocabulary detections, each with an identified object name, fused with UniDepth’s dense metric depth map (right), classroom photograph.

farthest) within the scene. Fig. 3 shows the corresponding annotated image alongside the underlying dense depth map.

As a quantitative sanity check that does not require ground truth, we test whether an object’s vertical position in the frame is consistently related to its predicted depth: in an ordinary photograph, an object whose bounding box extends further down the frame tends to be physically closer to the camera. The Spearman correlation between each detection’s lower box edge (y -coordinate) and its fused depth across all sixteen identified objects is $\rho = -0.76$ ($p = 0.0006$), a strong and statistically significant negative correlation, which is the expected sign and supports the claim that the fused distances are geometrically sensible rather than arbitrary. Table IV also shows this ordering directly: the six furniture objects nearest the camera (ranks 1–9) are exclusively chairs and desks, while the five farthest objects (ranks 12–16) are exclusively wall-mounted or structural elements, exactly the spatial layout a human would expect from a classroom photographed from the

back of the room.

D. Discussion

The final detector configuration produces sixteen clean, single-concept, correctly *named* labels with no repeated or merged label strings, a substantial improvement over both earlier attempts, and a 94% confident-identification rate overall. The spatial-consistency result is genuine evidence of geometric plausibility, though it is worth being precise about what it does and does not show: a strong rank correlation between frame position and depth is consistent with correct depth ordering, but it is not equivalent to a distance being correct in an absolute sense, since only the ordering, not the magnitude, is being tested. The one remaining unresolved detection is reported honestly rather than silently fixed, and is revisited in Section VIII.

VIII. LIMITATIONS

Several limitations apply across this study and are stated directly rather than left implicit. First, neither extension has access to ground-truth depth for our own photographs, so every quantitative result in Sections VI–VII is a self-referenced or detector-derived consistency measurement rather than an error against an external truth; this is an appropriate and clearly-labeled substitute given the constraint, not a claim of absolute accuracy. Second, Extension I was evaluated on a single classroom photograph; the degradation trends reported here are demonstrated, not proven to generalize, and repeating them across additional scenes is a natural next step. Third, the noise robustness results in Section VI use one random draw per severity level, which is sufficient to reveal the non-monotonic anomaly at $\sigma = 0.02$ but not sufficient to rule out sampling variance as its cause. Fourth, the object detector in Section VII is fundamentally a prompted, open-vocabulary system, not an unconstrained one: it can only report a label that was included, directly or through caption augmentation, in its vocabulary for that image, and one of seventeen candidate objects in the classroom photograph remains genuinely unresolved by this method. Finally, an informal observation made during reproduction is disclosed here rather than omitted: re-running inference on the same photograph across different Colab runtimes (a CPU session defaulting to UniDepthV2-ViT-S, a GPU session to UniDepthV2-ViT-L) produced visibly different absolute depth scales for the identical image. This study did not formally quantify that effect, since doing so rigorously falls outside the two extensions reported here, but the anecdote is a real, reproducible observation and a system relying on UniDepth for obstacle-distance thresholds should for this reason fix and log which backbone is loaded rather than allow it to vary silently with the available hardware; a controlled, quantified cross-backbone study is identified as a priority for future work below.

IX. CONCLUSION AND FUTURE WORK

This report reproduces UniDepth’s central claim, that metric depth and camera intrinsics can be recovered jointly from a

single uncalibrated RGB image, both qualitatively on two photographs of our own and quantitatively by validating the exact checkpoint used against its own published NYU-Depth-v2 and KITTI zero-shot numbers, where UniDepth-ViT-L reduces RMSE over the strongest published baseline by 40.4% on NYU-Depth-v2 and 22.6% on KITTI. Beyond reproduction, two extensions were carried out with fully quantified, non-fabricated results. The robustness study uncovers a sharp, non-gradual scale collapse under moderate low light – Agree@5% falling from 99.6% to 0.02% across a single severity step – together with a noise-severity anomaly worth further investigation. The object-level depth fusion pipeline required two rounds of detector replacement before producing sixteen clean, correctly named, geometrically consistent object distances at a 94% confident-identification rate, with a statistically significant spatial-consistency correlation ($\rho = -0.76$, $p = 0.0006$) confirming that the fused distances are geometrically sensible.

Future work follows directly from the limitations above: repeating the noise-robustness experiment across multiple random seeds per severity to separate genuine trend from sampling noise; extending both extensions to additional scenes beyond the single classroom photograph reported here; resolving the one outstanding unlabeled object in Extension II, potentially with a segmentation-based method rather than a bounding-box detector; converting the informal cross-runtime backbone-scale observation noted in Section VIII into a controlled, quantified study across all three officially released UniDepthV2 backbones; and, time permitting, acquiring a small set of tape-measure-verified reference distances for a handful of static objects in our own photographs so that at least a limited external ground-truth comparison becomes possible.

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